Esercizi - Linguaggi di programmazione

Determinare la grammatica che genera il seguente linguaggio:

1. $L = \{a^n \mid n>0\}$ = {a, aa, aaa, aaaa, ...}
2. $L = \{a^n \mid n>=0\}$ = {λ, a, aa, aaa, aaaa, ...}
3. $L = \{a^n \mid n=2h+1, h>0\}$ = {aaa, aaaaa, aaaaaaa, ...}
4. $L = \{a^n \mid n=2h+1, h>=0\}$ = {a, aaa, aaaaa, aaaaaaa, ...}
5. $L = \{a^n \mid n=2h, h>=0\}$ = {λ, aa, aaaa, aaaaaa, aaaaaaaa, ...}
6. $L = \{a^n \mid n=2h, h>0\}$ = {aa, aaaa, aaaaaa, aaaaaaaa, ...}
7. $L = \{a^nb^n \mid n>0\}$ = {ab, aabb, aaabbb, aaaabbbb, ...}
8. $L = \{a^nb^n \mid n>=0\}$ = {λ, ab, aabb, aaabbb, aaaabbbb, …}
9. $L = \{a^nb^{2n} \mid n>0\}$ = {ab, aabb, aaabbb, aaaabbbb, …}
10. $L = \{a^nb^{2n} \mid n>=0\}$ = {λ, abb, aabbbb, aaabbbbbb, aaaabbbbbbbb, …}
11. $L = \{w \mid w \in \{0, 1\}$, *w ha lo stesso numero di 0 e di 1*$\}$
12. $L = \{a^nb^k \mid n,k>0\}$ = {ab, abb, aab, aba, bab, aabb, abab, abbb, ...}
13. $L = \{a^nb^k \mid n,k>=0\}$ = { λ, a, b, ab, ba, aa, bb, abb, aab, aba, bab, aaa, bbb,...}
14. $L = \{a^nb^k \mid n>=0, k>0\}$ = {b, bb, ab, ba, bbb, abb, aab, aba, bab, aabb, abab, ...}
15. $L = \{a^nb^k \mid n>0, k>=0\}$ = {a, aa, ab, ba, aaa, abb, aab, aba, bab, aabb, abab, ...}
16. $L = \{a^nb^kc^{2n} \mid n,k>0\}$ = {abcc, abbcc, abbbcc, aabcccc, abbbbcc, …}
17. $L = \{a^nb^kc^{2n} \mid n,k>=0\}$ = {b, bb, acc, abcc, bbb, bbbb, abbcc, abbbcc, aabcccc, ..}
18. $L = \{a^nb^kc^{2n} \mid n>0, k>=0\}$ = {ac, abcc, abbcc, aacccc, abbbcc, abbbbcc,…}
19. $L = \{a^nb^kc^{3n} \mid n>0, k>=0\}$ = {accc, abccc, abbccc, aacccccc, abbbccc, abbbccc,…}
20. $L = \{a^nb^kc^{2n} \mid n>=0, k>0\}$ = {b, bb, bbb, abcc, bbbb, abbcc, abbbcc, aabcccc, …}
21. $L = \{a^nb^{2n+m}c^m \mid n>0, m>0\}$ = {abbbc, abbbbcc, abbbbbccc, aabbbbbc, …}
22. $L = \{a^nb^{2n+m}c^m \mid n=>0, m>0\}$ = {bc, bbcc, abbbc, bbbccc, …}
23. $L = \{a^nb^{2n+m}c^m \mid n>0, m=>0\}$ = {abb, aabbbb, abbbbcc, abbbbbccc, …}ì
24. $L = \{a^nb^{2n+m}c^m \mid n=>0, m=>0\}$ = {λ, bc, abb, bbcc, abbbc, aabbbb, bbbccc, …}
25. $L = \{a^nb^mc^{2n+m} \mid n>0, m>0\}$ = {abcc, aabbbc, abbbcc, aabbcccc, …}

    In autonomia: n=>0, m=>0, n>0, m=>0, n=>0, m>0.
26. $L = \{a^{2n+m}b^nc^m \mid n=>0, m=>0\}$ = {λ, ac, aab, aacc, aaabc, aaaccc, …}

    In autonomia: n>0, m>0, n>0, m=>0, n=>0, m>0.
27. $L = \{a^nb^kc^m \mid n>0, m>0, k>n+m\}$ = {abbbc, abbbbc, aabbbbc, aabbbbbcc, …}
28. $L = \{a^nb^kc^m \mid n=>0, m>0, k>n+m\}$ = {bbc, bbbc, abbbc, aabbb, abbbbc, …}
29. $L = \{a^nb^kc^m \mid n>0, m=>0, k>n+m\}$ = {abb, abbb, abbbc, abbbbc, aabbbbc, …}
30. $L = \{a^nb^kc^m \mid n>0, m>0, k=>n+m\}$ = {abc, abbc, abbbc, abbbbc, <u>aabbbc</u>, …}
31. $L = \{a^nb^kc^m \mid n=>0, m=>0, k>n+m\}$ = {b, bb, bbb, abbbc, bbbc, abbbbb, …}
32. $L = \{a^nb^kc^m \mid n=>0, m=>0, k=>n+m\}$ = {λ, b, bb, ab, bc, bbb, abbc, …}

33. $L = \{a^n b^k c^m \mid n=>0, m>0, k>n\}$ = {bc, bbc, bcc, bbcc, abbc, …}
34. $L = \{w \mid w \in \{a, b, c\}, w=aa\gamma, \gamma \in \{a, b, c\}^*\}$ *(tutte le stringhe che cominciano per aa)*
35. $L = \{(a^{2n}) \mid n=>0\}$ = {(), (aa), (aaaa), …}
36. $L = \{a^n b^m c^m d^n \mid n>0, m>0\}$ = {abcd, aabcdd, abbccd, aabbccdd, …}
37. $L = \{a^{2n} b^n \text{ oppure } b^{2n} a^n \mid n>0\}$ = {aab, bba, aaaabb, bbbbaa, …}
38. $L = \{\gamma^n c^i \mid n>0, i>0, \gamma = ab\}$ = {abc, abcc, abccc, ababc, …}
39. $L = \{wb^k w' \mid w,w' \in \{a, c\}^*, |w|+|w'|= k, k>0\}$ = {abbaa, aabbc, aabbbbc, …}
40. $L = \{wb^k w' \mid w,w' \in \{a, c\}^*, |w|+|w'|= 2k, k>0\}$ = {abba, cbbc, aabbbac, …}
41. $L = \{a^n b^k c^m \mid n=>0, m=>0, k=n \text{ oppure } k=2m+1\}$ = {λ, b, ab, aab, bbbc, aabb, …}
42. $L = \{a^n b^m \mid n=2i, m=2k, i>0, k>0\}$ *(a pari e b pari)* = {b, aabb, aaaabb, aabbbb, …}
43. $L = \{a^n b^m \mid n=2i, m=2k+1, i=>0, k>0\}$ *(a pari e b dispari)* = {b, aab, bbb, aaaab, …}
44. $L = \{a^n b^n c a^j b^j \mid i=>0, j>0\}$ = {cab, caabb, abcab, caaabbb, …}
45. $L = \{wcw^R \mid w,w' \in \{a, b\}^*, R = \text{palindromo}\}$ = {aca, abcba, aabcbaa, ….}

46. *Estendere la grammatica delle parentesi ben bilanciate aggiungendo anche la gestione delle parentesi graffe e quadre*